Uncertainty-Aware Emergency Route and ETA Prediction System

Final Year Machine Learning Project

1. Problem Statement

During emergencies, reaching a hospital quickly is very important. However, traffic jams, accidents, weather and road closures can make normal navigation estimates inaccurate. This project aims to develop a smart navigation system that predicts travel time, estimates the reliability of the prediction and suggests the best route to reach a hospital.

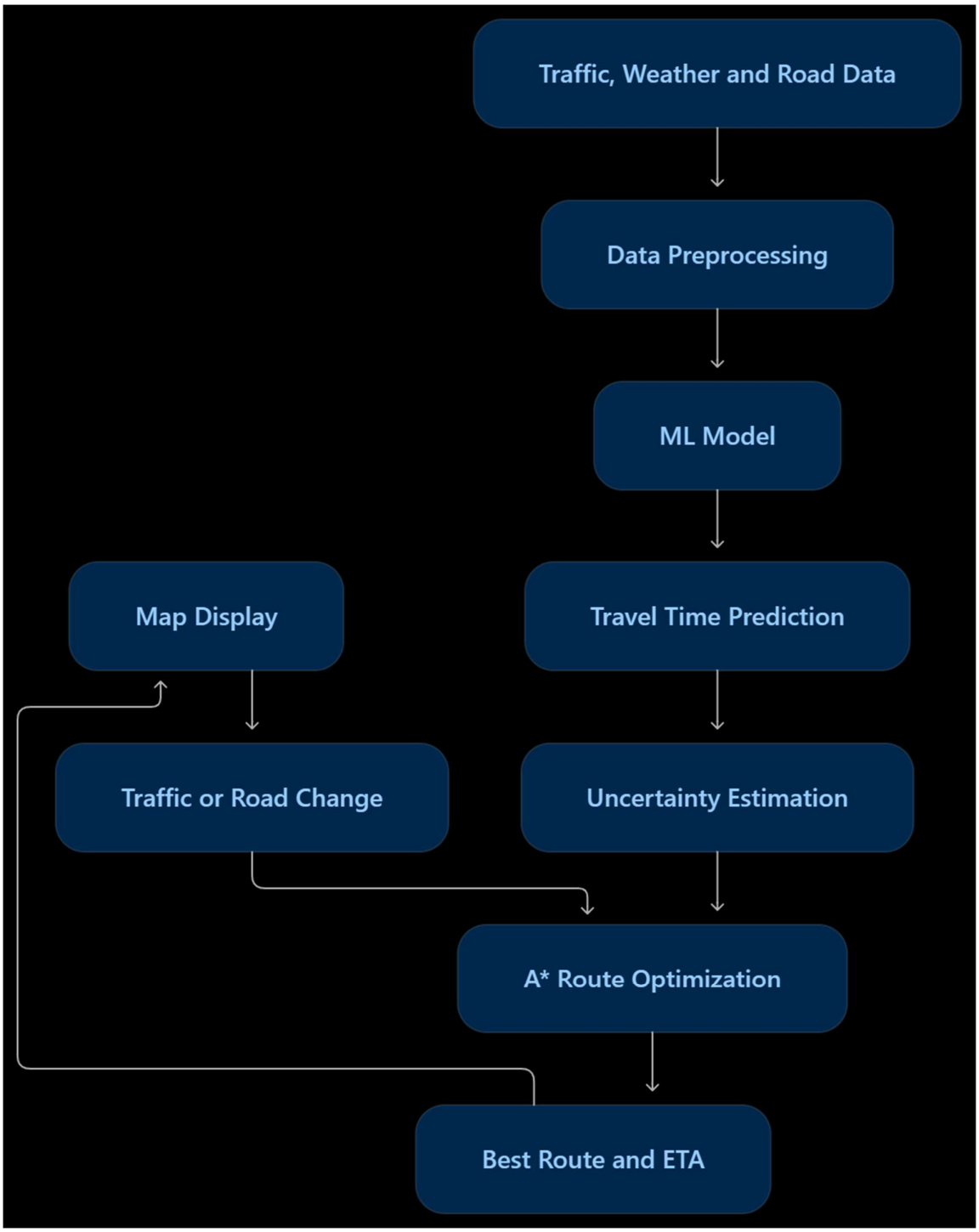
2. Introduction

The system combines Machine Learning and route optimization algorithms to improve emergency navigation. It uses road, traffic and weather data to predict travel time for different road segments. It also provides an estimated arrival-time range and finds the fastest or most reliable route. If road conditions change, the system automatically suggests an alternative route.

3. Main Objectives

1. Create a digital road network using OpenStreetMap.
2. Predict road travel time using Machine Learning.
3. Provide an estimated arrival time with a confidence range.
4. Find the fastest and most reliable route using A*.
5. Reroute automatically when traffic or road conditions change.
6. Suggest the best departure time.
7. Evaluate prediction accuracy using MAE and RMSE.

4. System Workflow



5. Technologies Used

Technology Stack – Emergency Route and ETA Prediction System

Here is the recommended tech stack for your final-year project.

Complete Tech Stack

Component	Technology	Purpose
Programming Language	Python	Main development language
ML Model	XGBoost / LightGBM	Predict road-segment travel time
Uncertainty Prediction	Quantile Regression	Predict ETA ranges
Road Network	OpenStreetMap (OSM)	Obtain road network data
Map Processing	OSMnx	Build road graphs
Route Optimization	A* Algorithm	Find fastest and reliable routes
Data Processing	Pandas, NumPy	Clean and process datasets
Visualization	Folium / Leaflet	Display routes on maps
Backend	FastAPI	Connect ML models with frontend
Frontend	React.js	Build the user interface
Database	PostgreSQL	Store road, prediction and route data
Model Evaluation	Scikit-learn	MAE, RMSE and model evaluation

Project References

1. Google DeepMind (2020), Traffic Prediction with Advanced Graph Neural Networks. Link

2. Zhao, Q. and Vanberkel, P. (2025), A Comparison of Ambulance Travel Time Approximation Using Google Maps and Machine Learning. Link

3. Ambulance Travel Time Estimation with Machine Learning (2024), Procedia Computer Science. Link

4. Beyond Expectation: Deep Joint Mean and Quantile Regression for Spatio-Temporal Problems (2018). Link

5. Boeing, G. (2017), OSMnx: New Methods for Acquiring, Constructing, Analyzing, and Visualizing Complex Street Networks. Link

6. Improving Estimation of Ambulance Travel Time Using Historical Transport and Weather Data. Link